Azure IoT Hub Reloaded: il presente che funziona, il futuro che arriva

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UNIPORDENONE
moltiplica il valore della formazione

TD SYNNEX

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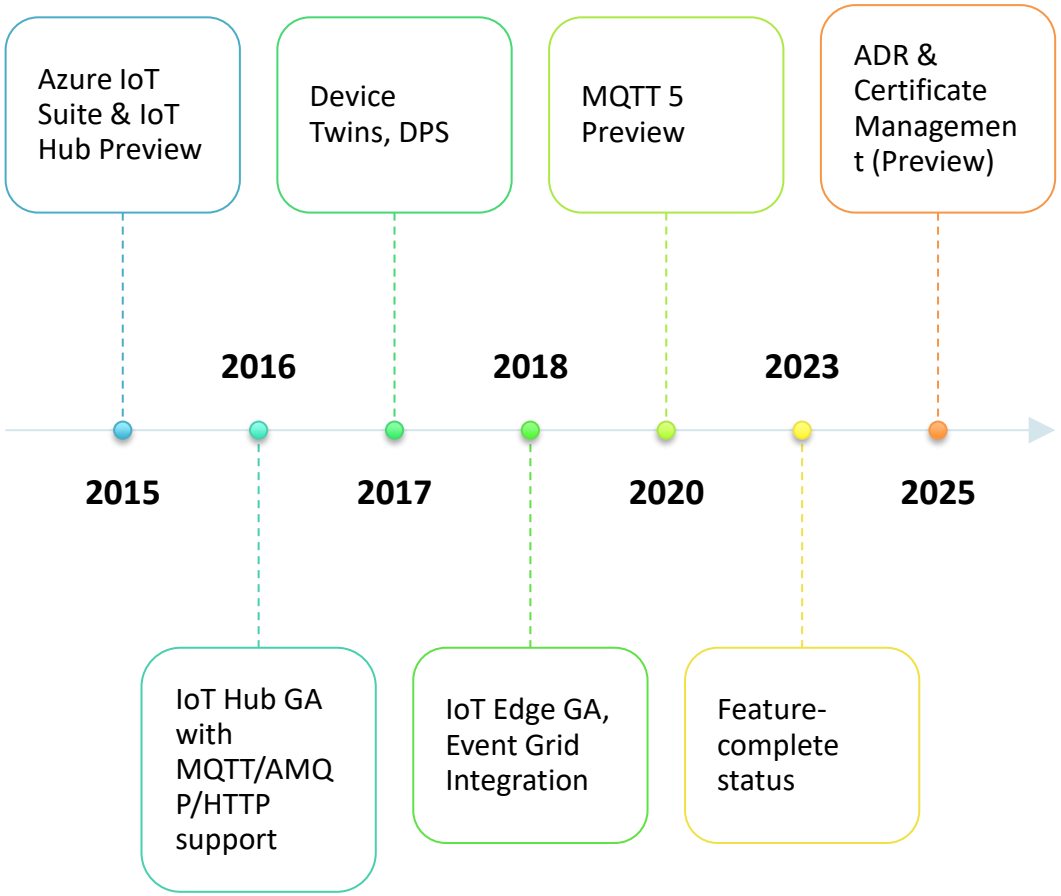
Azure IoT Hub: History & Evolution



IoT Hub Core Architecture

Device Identity Registry

Protocol Support: MQTT, AMQP, HTTP

Secure Access via SAS Tokens / X.509

Built-in endpoints and message routing

Integration with Azure services

Security Model

- Per-device authentication
- TLS encryption
- SAS tokens and X.509 certificates
- Integration with Defender for IoT
- Device Provisioning Service (DPS)

Messaging Patterns

DEVICE-TO-CLOUD TELEMETRY

CLOUD-TO-DEVICE COMMANDS

DIRECT METHODS

DEVICE TWINS

FILE UPLOADS

Device Management at Scale

- Device Twins: Desired & Reported Properties
- Device Tags for Grouping
- Jobs for Bulk Updates
- Azure Device Update Integration

Integration with Azure Services

- Message Routing to Storage, Event Hubs, etc.
- Event Grid for Device Events
- Stream Analytics, Digital Twins, Power BI
- Azure Functions & Logic Apps

Scalability & Reliability

- Millions of devices supported
- High throughput (events/sec)
- Built-in message retention (7 days)
- Monitoring via Azure Monitor
- SLA-backed availability

Recent Features & Previews

AZURE DEVICE REGISTRY INTEGRATION (PREVIEW)

X.509 CERTIFICATE MANAGEMENT (PREVIEW)

MQTT 5 PREVIEW (DEPRECATED)

DEFENDER FOR IOT INTEGRATION

IOT HUB FEATURE-COMPLETE STATUS

Azure IoT Operations Overview

- Edge-native IoT platform
- Runs on Arc-enabled Kubernetes
- Local MQTT Broker, OPC UA support
- Offline operation up to 72h
- Cloud-managed edge deployments

IoT Hub vs IoT Operations

IoT Hub: Cloud-first, PaaS

IoT Hub: Direct device-cloud connection

Complementary in hybrid IoT scenarios

IoT Ops: Edge-first, Kubernetes-based

IoT Ops: Local device connectivity

Complementary in hybrid IoT scenarios

When to Use What

- Use IoT Hub for cloud-managed devices
- Use IoT Ops for industrial edge scenarios
- Hybrid: Combine both for flexibility
- IoT Hub for consumer IoT
- IoT Ops for factory/plant deployments

Q&A

Questions and discussion

Thank you for attending!

- 30+ years in IT (Developer, Architect, Senior Consultant, PM, Trainer)
- Speaker, Community addicted
- IoT Influencer
- Microsoft Certified Trainer

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Feedback

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